



THE UNITED REPUBLIC OF TANZANIA

PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

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FORM FOUR ZONAL MOCK EXAMINATION

BASIC MATHEMATICS

YEAR 2026

TIME: 3:00 HOURS

INSTRUCTIONS

1. This paper consists of two sections **A** and **B** with a total of **fourteen (14)** questions
2. Answer **all** questions in both sections
3. All necessary working and answers for each question done must be shown clearly.
4. **NECTA** Mathematical tables and Non-programmable calculators may be used.
5. All communication devices and any unauthorized materials are **not allowed** in the examination room.
6. All writings should be in **blue** or **black** ink while drawings should be in **pencil**.
7. Write your **Examination Number** on every page of your answer booklet(s)
8. Where necessary use the following constants;
 - (i) Pie, $\pi = 3.14$
 - (ii) Radius of the Earth (R) = 6400 km

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SECTION A: (60 MARKS)

Answer all questions in this section

1. (a) A floor with length 500cm and 400cm width is to be covered with the largest size of the square tiles. If no space of the floor is to be left uncovered and the cost of each square tile is TZS 2500/=, how much will each cost to cover the floor?

(b) Write 356.476 correct to:
(i) Tenth
(ii) Two significant figures
2. (a) Solve for y is $2^{(y+7)} - 2^{(y+5)} = 3$
(b) By rationalizing the numerator, simplify the expression $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{5}+\sqrt{2}}$
3. (a) There are 100 form three students at Kabuku Secondary School; 68 students play football, 35 students play hockey and 10 students play neither game.
(i) Draw a Venn diagram to illustrate this information.
(ii) How many students play both football and hockey?
(iii) How many students play either football or hockey?

(b) John threw a coin and rolled a die simultaneously. What is probability of getting?
(i) A tail and a number greater than 3.
(ii) A head and a square number
4. (a) Find the equation of a line passing through the midpoint of the points $A(-3,2)$ and $B(1,-4)$ and which is perpendicular to line AB

(b) Given the vectors $a = 6i + 12j$, $b = -2i - j$ and $c = -6i + 3j$. Find
(i) $d = 3a - 2b + c$ (ii) $|3a + 2b - 2c|$
5. (a) An engineer was walking around a certain building which is in the shape of regular polygon. He determines that the measure of one exterior angle of the building is half of one hundred and twenty degrees.
(i) How many sides does the building have?
(ii) What is the name of the regular polygon the building have?

(b) The area of a regular octagon is 100cm^2 . Find the radius of the circle inscribing it.

6. (a) Lakunle has a pile of four books on his desk. The thickness of the books are 22mm, 13mm, 39mm and 18mm. Can the pile fit in the shelf of 11cm high? If so what is the length of the space left?
- (b) Baroka bought motor vehicle spare parts from Japan worth 109,000 Japanese Yen. When arrived in Tanzania was charged custom duty of 25% on spare parts. If the exchange rates were the following;
 1 US dollar = 2180 Japanese Yen
 1 US dollar = 2500 Tanzanian shillings. Calculate the duty he paid in Tanzanian shillings.
7. (a) At Mawenzi Village there are 2,000 people of whom 1,500 are adults. What the ratio of Adults to children in the village?
- (b) Mr. Kazimoto is an entrepreneur. He started a business on 1st January 2020 with a capital in cash of 2,000,000/= Tanzanian shillings.
- | | | |
|-----------|-------------------------|-----------------|
| January 2 | bought shelves for cash | 1,400,000/= Tsh |
| 3 | Sold goods for cash | 1,000,000/= Tsh |
| 6 | Paid rent for cash | 220,000/= Tsh |
| 26 | Paid wages for cash | 220,000/= Tsh |
| 30 | Sold goods for cash | 600,000/= Tsh |
- Required:** Prepare Cash Account and balance it.
8. (a) A small business man sells products worth 2,500,000/= Tanzanian shillings during his first year. He sets a goal of increasing annual sales by 550,000/= Tanzanian shillings each year. Assuming the goal met, find the total sales during the first 15 years of the business operation.
- (b) The first, second and fifth terms of arithmetic progressions are consecutive terms of geometric progressions. Find the possible values of the common ratio.
9. (a) Find the value of $\frac{\sin(-150^\circ) \cos 315^\circ}{\tan 300}$ without using trigonometrical table.
- (b) A rectangular plot of land is 40 metres long. If the length of its diagonal is 50metres, how wide is the plot?
10. (a) At a school game, student tickets were sold 50 shillings each and each adult ticket was sold twice the price of each student ticket. If the total receipts from 900 tickets were 50,000 shillings, how many tickets of each kind were sold?
- (b) Solve the equation $3x^2 - 15x + 6 = 0$ by completing the square method.

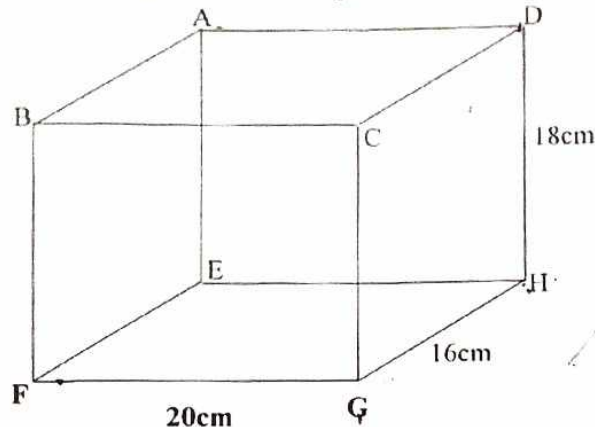
SECTION B: (40 MARKS)

Answer all questions

11. The table below shows the monthly test results for basic mathematics subject at Nyumbu Secondary School.

Marks	1-10	11-20	21-30	31-40	41-50	51-60	61-70
Frequency	8	7	12	5	4	4	10

- (i) State the modal class of the data given above.
 (ii) Find the mode and mean of the data. (Use assumed mean $A=36.5$)
 (iii) Draw the cumulative frequency curve and estimate median
12. (a) The following figure represents a rectangular prism in which $FG = 20\text{cm}$, $GH = 16\text{cm}$ and $HD = 18\text{cm}$.



Calculate:

- (i) The length $\overline{FD}$
 (ii) The angle between plane BGAH and the base EFGH.
 (iii) Its volume, give your answer in litres.
- (b) A bus leaves at Mbogwe stand ($5^\circ\text{S}, 40^\circ\text{E}$) at a constant speed of 20km/hr . how many hours will the bus take to reach Ushirombo stand ($15^\circ\text{S}, 40^\circ\text{E}$). Use $\pi = 3.14$ and the radius of the earth = 6400km
13. (a) Find the value of each unknown from the matrix equation

$$\begin{pmatrix} x & w \\ y & z \end{pmatrix} = A + B \quad \text{where } A = \begin{pmatrix} 3 & -2 \\ 1 & 4 \end{pmatrix} \text{ and } B = \begin{pmatrix} 6 & -5 \\ -4 & 7 \end{pmatrix}$$
- (b) Mbia is an entrepreneur who sells two types of fruits. One day weighed some fruits on scales. He found that 4 mangoes and 7 oranges weighed 450 grams while 6 mangoes and 5 oranges weighed 510 grams. Using suitable letters, write down two simultaneous equations. Hence use matrix method to find the total weight of 10 mangoes and 4 oranges.
14. (a) A function f is defined by $f(x) = \frac{2x+1}{x-3}$, find:
 (i) Domain and Range of $f(x)$.
 (ii) The value of $f(4)$
- (b) R & J contractors have 150 tons of sand and 90 tons of cement for building structures A and B for business. Structure A requires 1 tons of sand and 2 tons of cement whereas structure B requires 3 tons of sand and 1 tons of cement. If structure A is sold at a profit of shs 400,000 and structure B at a profit of shs 600,000. How many structures of each type should be made in order to get maximum profit?